Expert Information

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First Name

Last Name

Email

Position Title

Organization / University

Instructions

Purpose

The purpose of this research is to develop and validate the content for a new **Quantum Information Science Concept Introductory Test (QISCIT)**.

As implied by its name, QISCIT focuses on introductory QIS concepts that can be demonstrated without the formalism or linear algebra required by other quantum assessments. QISCIT has been designed to focus on the following quantum concepts: *superposition*, *measurement*, *gate operations*, *entanglement*, *teleportation*, and *decoherence*.

We are employing the process of Almanasreh et al. (https://doi.org/10.1016/j.sapharm.2018.03.066) to evaluate the content validity of QISCIT. This process consists of **two rounds**: 1) quantifying the content validity of each item and the initial overall instrument based on ratings from experts, such as yourself, to determine which items to delete, modify, or replace, and 2) quantifying the content validity of each revised item and the final overall instrument. Your estimated commitment is a **total of 2 hours**: 1 hour for the first round and 1 hour for the second round.

The development and validation of QISCIT is funded by the National Science Foundation (NSF) Award #2302816: "Collaborative Research: Advancing Quantum Education by Adaptively Addressing Misconceptions in Virtual Reality".

Instructions

On the following pages, you will be presented with QISCIT, which includes a glossary of QIS concepts and a total of 22 questions grouped by primary concept for your convenience. For each item, we ask that you **rate** the item based on content relevance from 1 ("Not Relevant") to 4 ("Extremely Relevant") and **provide comments** and make suggestions regarding the wording of the item. Finally, you will also have the opportunity to **suggest additional items** to be added to QISCIT for consideration in round 2.

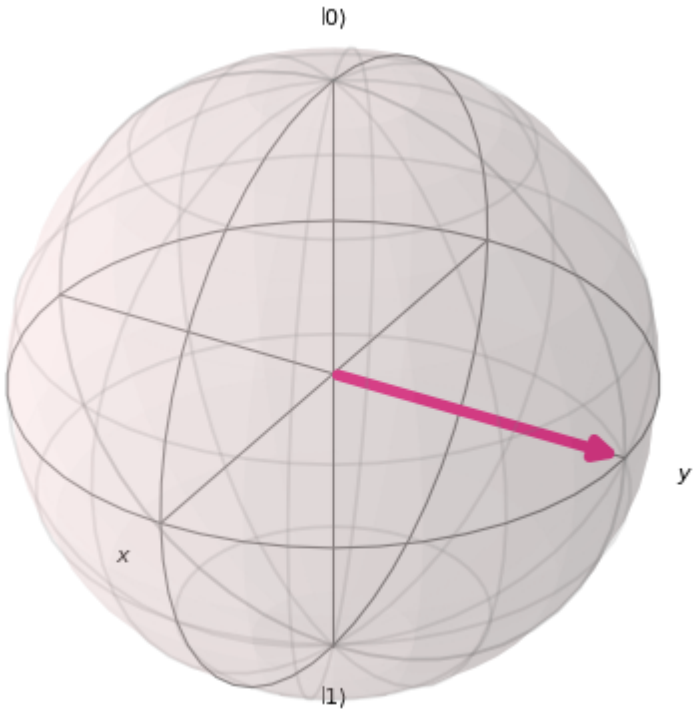
Glossary

The following glossary will be available on each page of questions. Please take time to briefly read the glossary and answer the following attention check question.

We will ask you to give feedback on the glossary, which will be available to students throughout the assessment. Note that certain concepts have been defined restrictively, such as defining measurement to be in the computational basis, to reduce cognitive load on students.

Glossary of Quantum Information Science Concepts

- **Quantum Bit** (a.k.a. **Qubit**): The basic unit of quantum information. It is the quantum analogue of a bit, with two basis states 0 and 1.
- **Superposition**: The ability of a quantum system to be in multiple basis states at the same time.
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- **Bloch Vector:** A method of visualizing the state of a qubit by converting its average measurement outcome to a point on or within a sphere. A picture of a Bloch vector is shown below.



- **Gate**: A quantum operation that can change the state of one or more qubits.
- **X Gate**: A quantum gate that rotates the Bloch vector by 180 degrees about the X axis.
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Please rate the glossary above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the glossary above.

How many degrees does the T gate rotate a qubit's Bloch vector around the Z axis?

45 degrees

90 degrees

180 degrees

The T gate does not rotate a qubit about the Z axis

Please rate the item above based on content relevance.

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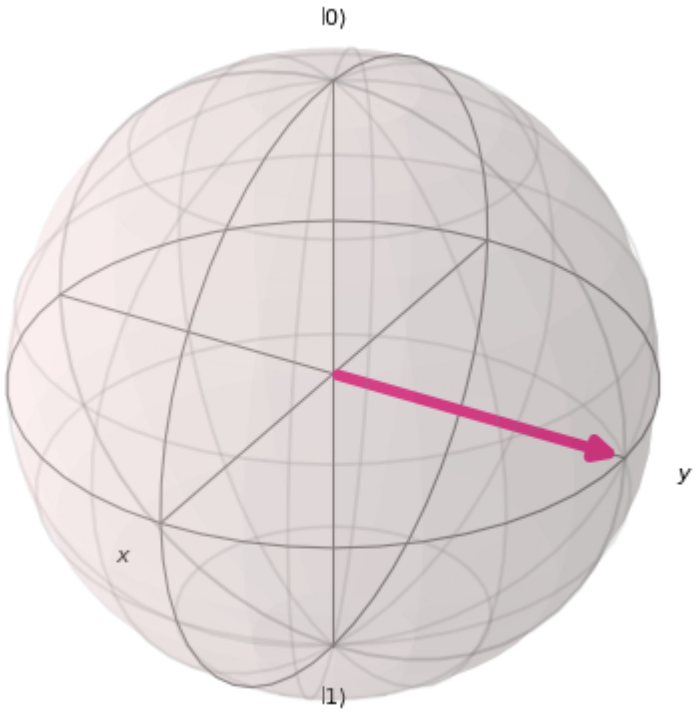
QISCIT

QIS Introductory Concept Test

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Question #1 (Superposition)

What does it mean for a single isolated qubit to be in a superposition of states 0 and 1?

When measured, the qubit will be in both state 0 and state 1
When measured, the qubit will be in either state 0 or 1
When measured, the qubit will be in neither state 0 nor 1
When measured, the qubit will remain in state 0 or state 1

Please rate the item above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #2 (Superposition)

How many mutually exclusive states exist for a single isolated qubit?

One

Two

Three

Infinity

Please rate the item above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #3 (Superposition)

How many potential superpositions exist for a single qubit?

One

Two

Three

Infinity

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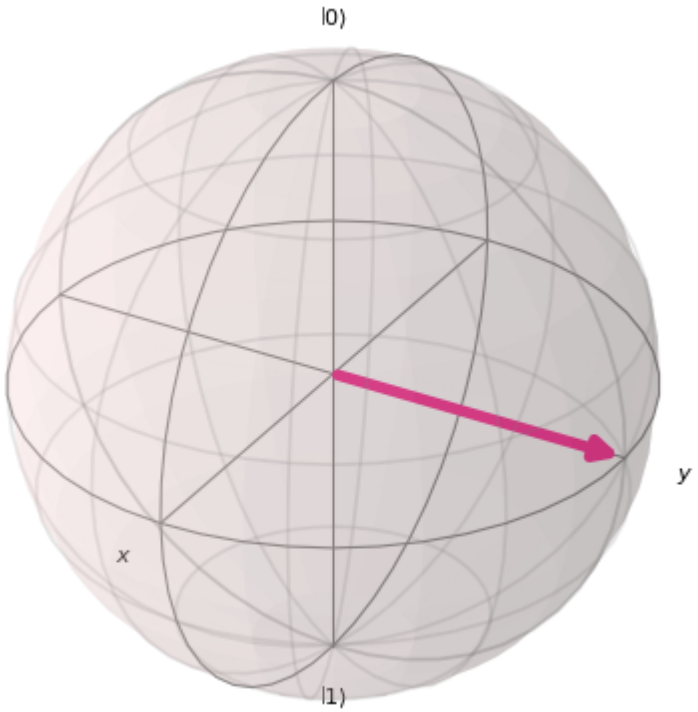
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Question #4 (Measurement)

How many possible measurement outcomes does a single qubit in a superposition state have?

One

Two

Three

Infinity

Please rate the item above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #5 (Measurement)

How many possible measurement outcomes does a single qubit have immediately after a

prior measurement in the same basis?

One

Two

Three

Infinity

Please rate the item above based on content relevance.

1 - Not Relevant 2 3 4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #6 (Measurement)

When are measurements not reversible?

Measurements are always reversible

Measurements are never reversible

Only when the qubits are entangled

Only when the qubits are in superpositions

Please rate the item above based on content relevance.

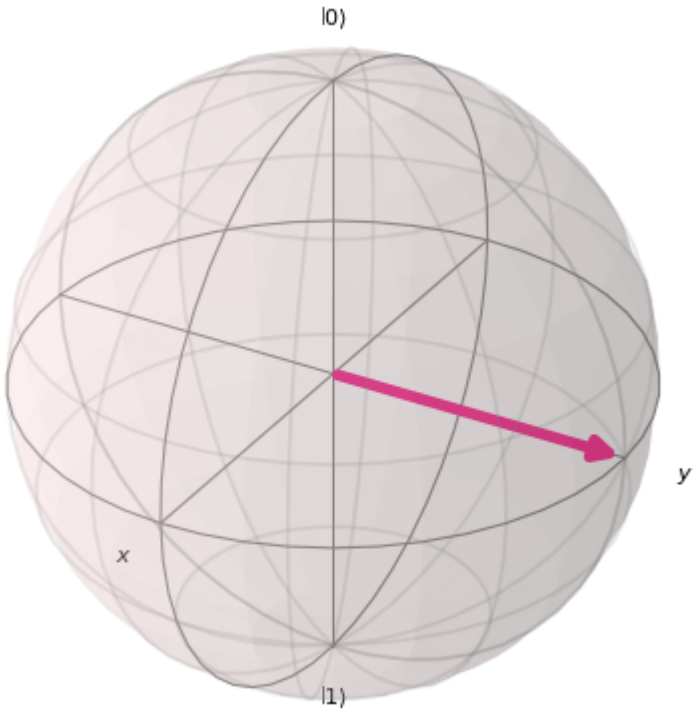
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Question #7 (Gates)

When are quantum gates not reversible?

Only when the qubits are entangled
Only when the qubits are in superpositions
Quantum gates are always reversible
Quantum gates are never reversible

Please rate the item above based on content relevance.

1 - Not Relevant | 2 | 3 | 4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #8 (Gates)

How can the operation of an X gate be reversed?

Apply an H gate

Apply an S gate

Apply a T gate

Apply an X gate

Please rate the item above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #9 (Gates)

A qubit starts in state 0 and then has an H gate applied to it. What is the probability of the qubit being in state 0 when measured?

0%

25%

50%

75%

100%

Please rate the item above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #10 (Gates)

How can a qubit be put into superposition starting from state 0?

Apply an H gate

Apply an S gate

Apply a T gate

Apply an X gate

Please rate the item above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #11 (Gates)

How can the operation of an S gate be reversed?

Apply another S gate

Apply another two S gates

Apply another three S gates

The S gate cannot be reversed

Please rate the item above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #12 (Gates)

How can the operations of two S gates be reversed?

Apply a T gate

Apply two T gates

Apply four T gates

The S gates cannot be reversed

Please rate the item above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #13 (Gates)

Two qubits each start in state 1. A CNOT gate is applied, using the first qubit as the control and the second qubit as the target. What is the probability of the second qubit being in state 0?

0%
25%
50%
75%
100%

Please rate the item above based on content relevance.

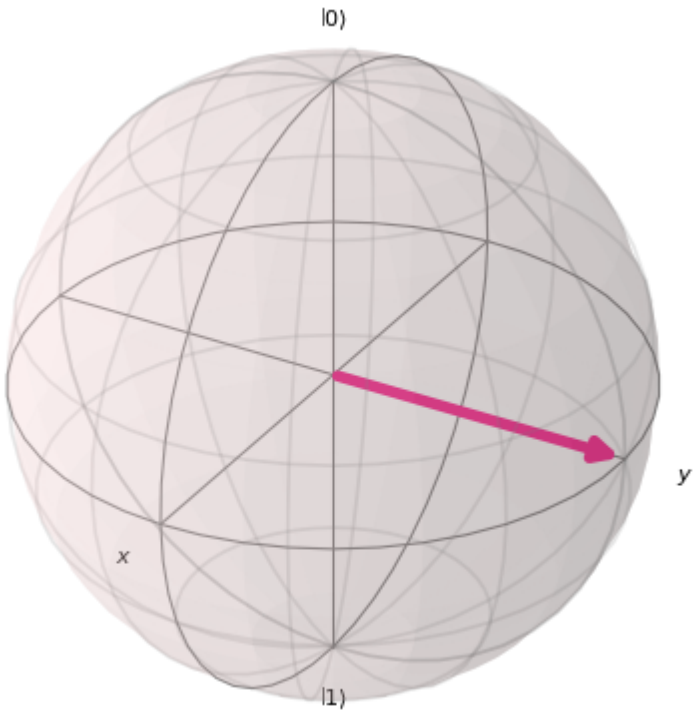
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Question #14 (Entanglement)

Two qubits are entangled, and then the first qubit is measured. What happened to the entanglement between the two qubits?

The entanglement increased

The entanglement remained the same

The entanglement depends on the result of the measurement

The entanglement decreased to zero

Please rate the item above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #15 (Entanglement)

Two isolated qubits are entangled. What can we say about the first qubit's Bloch vector?

It has length one

It has length less than one

It has length zero

It is opposite the second qubit

Please rate the item above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #16 (Entanglement)

Two qubits each start in state 1. An H gate is applied to the first qubit. A CNOT gate is then applied, using the first qubit as the control and the second qubit as the target. What

is the probability of the second qubit being in state 0?

0%
25%
50%
75%
100%

Please rate the item above based on content relevance.

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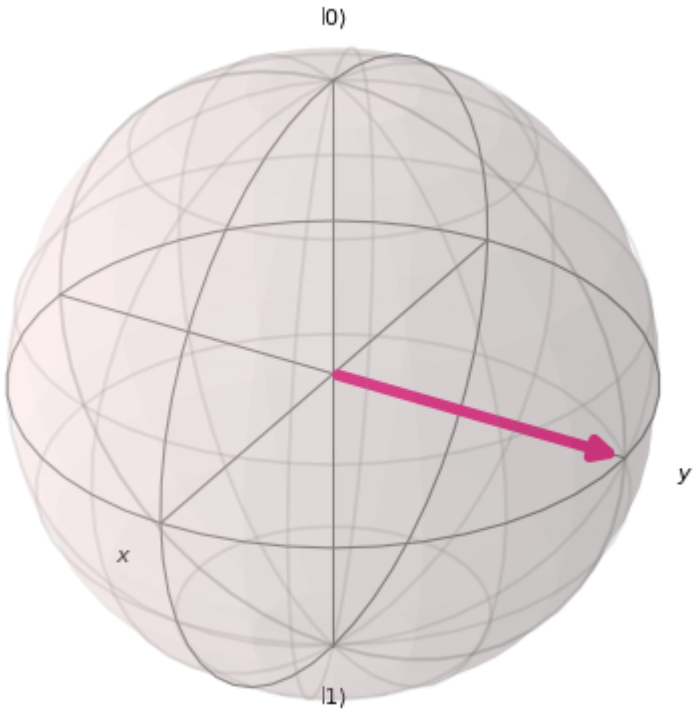
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Question #17 (Teleportation)

What does quantum teleportation transfer from one location to another?

A specific single-qubit state

A specific two-qubit state

Any single-qubit state

Any single classical state

Please rate the item above based on content relevance.

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Please provide any comments or suggestions regarding the wording of the item above.

Question #18 (Teleportation)

What is one non-local resource that is required for quantum teleportation?

A non-locally entangled pair of qubits

Non-local quantum gates

Non-local quantum measurements

There are no non-local resources required

Please rate the item above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #19 (Teleportation)

What channels are required for quantum teleportation?

One classical channel

One classical channel and one quantum channel

One quantum channel

Two quantum channels

Please rate the item above based on content relevance.

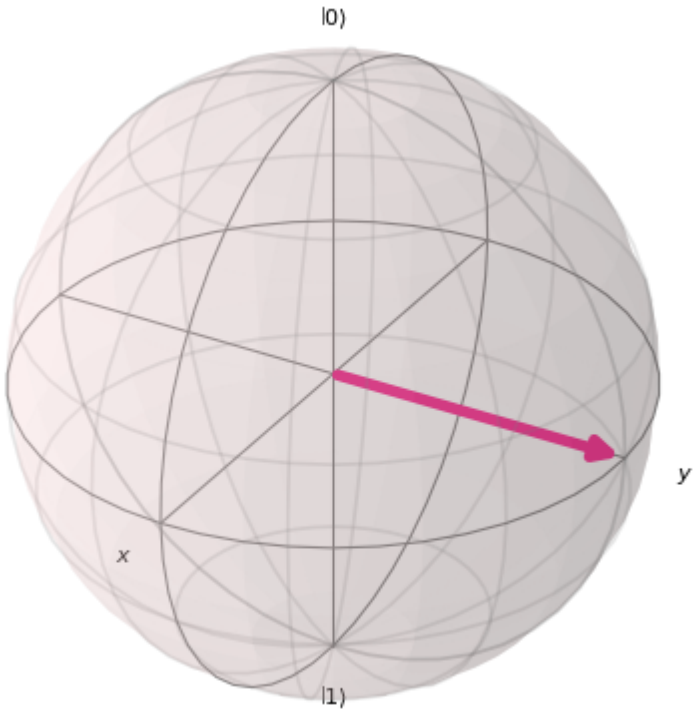
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Question #20 (Decoherence)

When is decoherence not reversible?

Decoherence is always reversible

Decoherence is never reversible

Only when the qubits are entangled

Only when the qubits are in superpositions

Please rate the item above based on content relevance.

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Please provide any comments or suggestions regarding the wording of the item above.

Question #21 (Decoherence)

Quantum decoherence involves the loss of quantum information due to what?

Qubits becoming entangled with one another

Qubits being measured

Qubits having quantum gates applied to them

Qubits interacting with the environment

Please rate the item above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #22 (Decoherence)

Quantum error correction involves the mitigation of errors due to what?

Decoherence

Entanglement

Gates

Measurement

Please rate the item above based on content relevance.

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Please provide any comments or suggestions regarding the wording of the item above.

Additional Feedback

Are there any relevant QIS concepts not addressed by the previous items?

Are there any QIS misconceptions that you have encountered that are not addressed by the previous items?

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